

Keli Liu

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Research Interests: Edge Computing, Microservice Deployment, Resource Optimization, Wireless Sensing

EDUCATION

Guizhou University, Guiyang, China

09/2023 - 06/2026

Master of Engineering in Electronic Information, State Key Laboratory of Public Big Data, Supervisor: Prof. Xiaoli Ruan; co-supervisor: Prof. Jing Yang.

- **GPA:** 3.53/4.00
- **Award:** Postgraduate National Scholarship (2/53, top 4%, 2025); Postgraduate Academic Scholarship (2023-2024, 2025-2026)
- **Core Modules:** Data Mining and Its Applications(95/100), Artificial Intelligence and Machine Learning(90/100), Big Data Technology and Applications (92/100), Statistics and Data Analysis Foundation (87/100)

Henan Finance University, Zhengzhou, China

09/2018 - 06/2022

Bachelor of Engineering in Software Engineering, School of Software Engineering

- **GPA:** 3.53/5.00
- **Honors:** Merit Student (2018-2019), Science and Technology Innovation Scholarship (2020-2021)
- **Core Modules:** Linux Operating System Foundation(90/100), Software Quality Assurance and Testing(91/100), Probability Theory and Mathematical Statistics(91/100), Web Design and Production(86/100)

RESEARCH INTERESTS

Edge Computing: Microservice Deployment, Task Offloading, Resource Optimization, Reinforcement Learning

Smart Sensing: Wireless Sensing, IOT hardware, Mobile Health

PUBLICATIONS

- **Keli Liu**, Jing Yang, et al. "SAMD-SRG: Service-Aware Microservice Deployment Using Spectral Ranking and Graph-Enhanced Reinforcement Learning." IEEE Transactions on Cognitive Communications and Networking.(**IEEE TCCN**, **IF= 7.0**, **JCR Q1**). [Accepted] <https://ieeexplore.ieee.org/document/11127211>
- **Keli Liu**, Jing Yang, et al. "TFRL-MD: Transformer-Enhanced Federated Reinforcement Learning for Intelligent Microservice Deployment and Resource Allocation." IEEE Transactions on Cloud Computing.(**IEEE TCC**, **CCF B**, **IF= 5.3**, **JCR Q1**). [Accepted] <https://ieeexplore.ieee.org/abstract/document/11563616>
- **Keli Liu**, Jing Yang, et al. "GraphSAGE-Enhanced Reinforcement Learning for Optimizing Load-Aware Microservice Deployment." The 3rd International Conference on Artificial Intelligence of Things and Systems.(**AIoTSys 2025**). [Accepted] https://doi.org/10.1007/978-981-95-2581-2_23
- **Keli Liu**, Jing Yang, et al. "PLMD-GRL: Proximity-Aware and Load-Balanced Microservice Deployment for Graph Reinforcement Learning in Edge-Cloud Environments." IEEE Transactions on Network Science and Engineering.(**IEEE TNSE**, **IF= 7.3**, **JCR Q1**). [Under Review]
- Jia Zhao, Shaobo Li, **Keli Liu**, et al. "TQMO: Trajectory-Aware QoE-Driven Intelligent Task Offloading for the Vehicular Metaverse." IEEE Internet of Things Journal.(**IEEE IoTJ**, **IF= 8.9**, **JCR Q1**). [Under Review]
- Zhaohu Zhang, Jing Yang, Xiaoli Ruan, **Keli Liu**, et al. "TEMA-PPO: Two-Stage Evolution-Enhanced Multi-Agent Proximal Policy Optimization for Latency-Optimal Task Scheduling in Edge Computing." IEEE Internet of Things Journal.(**IEEE IoTJ**, **IF= 8.9**, **JCR Q1**). [Under Review]

PATENTS

- Jing Yang, **Keli Liu**, et al. "Microservice Deployment Method Based on Proximity Awareness and Load Balancing." Invention Patent, CN202510109993.4, **Authorized**. (Student First Inventor)
- Jing Yang, Yang Zhao, Xiaoli Ruan, **Keli Liu**, et al. "Active Prediction Method for Ultra-Dense Network CDS Hotspots Considering Spatiotemporal Information." Invention Patent, CN202510255631.6, **Authorized**. (Student Second Inventor)

RESEARCH EXPERIENCE

Intelligent Perception and Measurement Technology for Computing Networks

07/2024 - 04/2026

Core Member, Guizhou Provincial Science and Technology Program

Guiyang, China

- Wrote project proposals and formulated resource performance optimization solutions tailored for computing network scenarios

- Developed a distributed federated reinforcement learning architecture, enabling privacy-preserving collaborative learning across edge nodes for optimal deployment strategies
- **Outcome:** One paper accepted by IEEE Transactions on Cloud Computing (IEEE TCC)

Computing Power Trading Technology for New Domestic Computing Centers

07/2024 - 04/2025

Core Member, Guizhou Provincial Science and Technology Program

Guiyang, China

- Proposed GNN-based context abstraction to analyze coupling relationships between deployment decisions and heterogeneous resource costs, achieving joint optimization in domestic computing power trading scenarios
- Drafted project proposals and defined technical architecture for computing power trading systems
- **Outcome:** One paper accepted by IEEE Transactions on Cognitive Communications and Networking (IEEE TCCN)

Massive Data Acceleration Transmission Methods for East-West Computing Scenarios

07/2024 - 01/2025

Core Member, Guizhou Provincial Science and Technology Program

Guiyang, China

- Designed a graph reinforcement learning method to analyze feature representations of task dependency relationships for optimized data transmission scheduling
- Proposed an affinity-aware method to optimize geographical deployment of microservices, reducing inter-node communication latency
- **Outcome:** Submitted one paper to IEEE TNSE, currently under review. One patent has been authorized

Adaptive Traffic Scheduling Technology with Offloading Delay Awareness

09/2023 - 05/2025

Core Member, Guizhou Provincial Science and Technology Program

Guiyang, China

- Developed adaptive scheduling framework based on deep reinforcement learning with offloading delay awareness, achieving reduced response latency in edge computing scenarios
- **Outcome:** One paper accepted by AIOTSys 2025

Performance Evaluation Methods for Large-Scale Edge Server Equipment under Multi-Service Mixed Operation Mode

09/2023 - 12/2024

Group Member, Guizhou Baishan Cloud Technology Co., Ltd.

Guiyang, China

- Researched performance evaluation methods for large-scale edge server equipment under multi-service mixed operation modes
- Developed algorithmic models for large-scale edge server clusters to improve prediction accuracy
- **Outcome:** Wrote Business Requirements Specification and Software Project Plan documents

HONORS & AWARDS

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| • 3rd Prize , 2025 (11th) National College Student Statistical Modeling Competition | Provincial Level, 08/2025 |
| • 3rd Prize , China International College Students' Innovation Competition | School Level, 06/2025 |
| • 2nd Prize , "HUAWEI Cup" The 21st China Post-Graduate Mathematical Contest in Modeling | National Level , 12/2024 |
| • 2nd Prize , China International College Students' Innovation Competition | School Level, 09/2024 |
| • 2nd Prize , China International College Students' Innovation Competition | School Level, 09/2024 |
| • 3rd Prize , China International College Students' Innovation Competition | Provincial Level, 08/2024 |
| • 3rd Prize , 2024 (10th) National College Student Statistical Modeling Competition | Provincial Level, 07/2024 |
| • 1st Prize , The 14th "Challenge Cup" National College Student Business Plan Competition | School Level, 06/2024 |
| • 2nd Prize , "Houpu Cup" Business Website Design Competition | School Level, 12/2018 |

SKILLS

Languages: Chinese (native), English (CET-6, IELTS 6.5)

Programming Languages: Python (reinforcement learning, deep learning, data analysis), Java, C

Software and Tools: MATLAB, Origin, Microsoft Office Suite, Adobe Photoshop